



Universität
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Blockchain & DLT
Research Group



Network Science HS24

Assignment 1

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FOR STUDIES PURPOSES ONLY

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Solution

1. Diameter

$$d_{max} = \max_{i,j \in V} d_{ij} = d_{1,9} = 8 \quad (1)$$

2. Distance between Two Nodes in a Chain

Given a chain $C = (V, E)$ of length $n - 1$, by definition 1:

$$E = \{(i, i + 1), i = 1, \dots, n - 1\}$$

$$V = \{1, \dots, n\}$$

Further, d_{ij} is equivalent to the shortest path between vertices i and j .

Given the one-dimensional linear nature of chains, d_{ij} is given by the consecutive edges between i and j :

$$d_{ij} = |i - j|$$

3. Closed Form for Average Distance of a Chain C with Odd n

The average distance $\langle d \rangle$ for a chain C , using definitions from ex.2, is defined as:

$$\langle d \rangle = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n d_{ij} = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n |i - j|$$

$\sum_{i=1}^n \sum_{j=1}^n |i - j|$ is the sum of all shortest paths in C . We can leverage the symmetric property ($d_{ij} \equiv d_{ji}$) of C to simplify the summation and eliminate the absolute value term:

$$\langle d \rangle = \frac{1}{n(n-1)} 2 * \sum_{i=1}^n \sum_{j=i+1}^n (j - i)$$

Intuitively this can be understood as summing up all shortest paths for $i < j$. The lengths of all shortest paths for $j < i$ is equivalent to that of $i < j$, since C is symmetric. I.e., whether we look ahead from the left or look back from the right is equivalent.

All that remains is calculating the closed form for $\langle d \rangle$

1. Inner Sum

$$\sum_{j=i+1}^n j - i = \sum_{i=1}^n j - \sum_{j=i+1}^n i = \frac{n(n+1)}{2} - \frac{i(i+1)}{2} - ni + i^2$$

2. Outer Sum

$$\sum_{i=1}^n \frac{n(n+1)}{2} - \frac{i(i+1)}{2} - ni + i^2$$

$$\begin{aligned}
&= n \frac{n(n+1)}{2} - \frac{1}{2} \left(\sum_{i=1}^n i^2 + \sum_{i=1}^n i \right) - n \sum_{i=1}^n i + \sum_{i=1}^n i^2 \\
&= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)(2n+1)}{12} - \frac{n(n+1)}{4} \\
&= \frac{n(n+1)(2n+1) - 3n(n+1)}{12} \\
&= \frac{n^3 - n}{6}
\end{aligned}$$

3. Symmetric Property of Chain (*2)

By the symmetric property, as outlined above, the total of all paths is double that of the double sum.

$$\frac{n^3 - n}{3}$$

4. Average

The average as defined in equation 1:

$$\begin{aligned}
&\frac{1}{n(n-1)} * \frac{n^3 - n}{3} \\
\langle d \rangle &= \frac{n^2 - 1}{3(n-1)} = \frac{n+1}{3}
\end{aligned}$$

4. Calculate the Average Distance for given Ring R

For the given ring R with even n , we need to determine d_{ij} as defined in the average distance in:

$$\langle d \rangle = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n d_{ij}$$

d_{ij} in a ring $R(V, E)$ with even n is given as:

$$d_{ij} = \begin{cases} |i - j| & \text{if } |i - j| \leq \frac{n}{2} \\ n - |i - j| & \text{else} \end{cases}$$

, since the shortest path for all $|i - j| \leq \frac{n}{2}$ we 'traverse' the ring in direction i to j , and for all $|i - j| > \frac{n}{2}$ from j to i reversed.

Assuming arbitrary node i . For $d = |i - j| \in \{1, \dots, \frac{n}{2}\}$ we get $\sum_{d=1}^{n/2} d$. For $d = |i - j| \in \{\frac{n}{2} + 1, \dots, n\}$ all except $\frac{n}{2}$ repeat, owed to the symmetry of ring R , which implies:

$$2 * \sum_{d=1}^{\frac{n}{2}} d - \frac{n}{2}$$

The average distance for node i thus yields:

$$\frac{1}{n-1} * 2 * \sum_{d=1}^{\frac{n}{2}} d - \frac{n}{2} = \frac{(\frac{n}{2})^2}{n-1}$$

The full graph's average distance $\langle d \rangle$ is equivalent to that of a single node, since all nodes are connected with the same properties. It follows:

$$\langle d_i \rangle = \frac{(\frac{n}{2})^2}{n-1} = \langle d \rangle = \frac{36}{11} = 3.27$$

, for the given ring in figure 3.

5. Explicit Formula for a ring $R(V, E)$ with odd n

Using the findings from exercise 4. we can directly deduce that for odd n :

$$d_{ij} = \begin{cases} |i-j| & \text{if } |i-j| < \frac{n}{2} \\ n - |i-j| & \text{else} \end{cases}$$

Assuming arbitrary node i . For all $d = |i-j| \in \{1, \dots, \lfloor \frac{n}{2} \rfloor\}$ we get $\sum_{d=1}^{(n-1)/2} d$.

The ring is again symmetric, thus including $|i-j| \in \{\lceil \frac{n}{2} \rceil, \dots, n\}$ the average distance for arbitrary node i :

$$L_i = \frac{1}{n-1} * 2 * \sum_{d=1}^{\frac{n-1}{2}} d = \frac{1}{n-1} * 2 * \frac{(\frac{n-1}{2})(\frac{n-1}{2} + 1)}{2} = \frac{1}{n-1} \frac{(n-1)(n+1)}{4}$$

$$L_1 = \frac{n+1}{4}$$

Leveraging the same property as in the exercise before, namely that the average distance L_i is equivalent with $\langle d \rangle$, we get:

$$\langle d \rangle = \frac{n+1}{4}$$

6. Closed Form for Average Distance of a 1-d Lattice with odd n and $n-1$ divisible by $2k$

Given a 1-d lattice with odd n and coordination number k . Since $n-1$ is divisible by $2k$, $\frac{n-1}{2}$ is divisible by k . Further the largest distance exhibited, i.e., the tipping point owed to the symmetry of an odd n 1-d lattice, is at $\frac{n-1}{2k}$, thus:

$$d_{ij} = \begin{cases} \lceil \frac{|i-j|}{k} \rceil & \text{if } |i-j| < \frac{n}{2} \\ \lceil \frac{n - |i-j|}{k} \rceil & \text{else} \end{cases}$$

The total distance td_i for arbitrary node i in a 1-d lattice with even n and $d = |i-j|$ yields:

$$td_i = 2 * \left(\sum_{d=1}^{\frac{n-1}{2k}} d * k \right) = 2k * \frac{\frac{n-1}{2k} \left(\frac{n-1}{2k} + 1 \right)}{2} = \frac{(n-1)(n-1+2k)}{4k}$$

The sum's upper bound stems from the maximum distance from node i to any node j , and since $\frac{n-1}{2}$ is divisible by k , each distance $d \in \{1, \dots, (\frac{n-1}{2k})\}$ is repeated k times. To account for both sides of the symmetry we multiply by 2.

Consequently, the average distance for a node i and again the full lattice is therefore:

$$\frac{1}{n-1} td_i = L_i \equiv \langle d \rangle = \frac{n-1+2k}{4k}$$